

Q1 2025 Post Earnings Analyst Call

Company Participants

- Manish Bhatia, Executive Vice President, Global Operations
- Mark Murphy, Executive Vice President and Chief Financial Officer
- Satya Kumar, Corporate Vice President, Investor Relations and Treasury
- Sumit Sadana, Executive Vice President and Chief Business Officer

Other Participants

- Aaron Rakers, Wells Fargo
- Brian Chin, Stifel Financial Corp.
- Chris Caso, Wolfe Research
- Hadi Orabi, TD Cowen
- Harlan Sur, JPMorgan Securities LLC
- Quinn Bolton, Needham & Company
- Timothy Arcuri, UBS
- Toshiya Hari, Goldman Sachs
- Vijay Rakesh, Mizuho Securities

Presentation

Operator

Thank you for standing by and welcome to Micron's Post Earnings Analyst Call. At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions). As a reminder, today's program is being recorded.

And now I'd like to introduce your host for today's program, Satya Kumar, Corporate Vice President, Investor Relations and Treasury. Please go ahead, sir.

Satya Kumar {[BIO](#) [18241786](#) <GO>}

Yes, thank you, Jonathan, and thank you, and welcome to Micron Technology's fiscal first quarter 2025 post-earnings analyst call. Joining me today are Sumit Sadana, Micron's Chief Business Officer; Manish Bhatia, EVP of Global Operations; and Mark Murphy, our CFO.

As a reminder, the matters we're discussing today include forward-looking statements regarding market demand and supply, market trends and drivers, and our expected results and guidance and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today.

We refer you to documents we have filed with the SEC, including our most recent Form 10-Q and upcoming 10-Q, for a discussion of risks that may affect our results. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, and achievements. We are under no duty to update any of the forward-looking statements to confirm these statements to actual results.

We can now open the call to Q&A.

Questions And Answers

Operator

(Question And Answer)

Certainly. And our first question for today comes from the line of Harlan Sur from JPMorgan. Your question, please.

Q - Harlan Sur {BIO 6539622 <GO>}

Yes. Thanks, guys for hosting this call back, so I've got two questions. On the 100 basis point decline in gross margins in Q2, as you guys mentioned, it's almost all NAND related, NICs, pricing, lower shipments, but qualitatively, like, what are you guys anticipating for DRAM pricing, right? You talked about positive mix impacts, HBM, server DRAM, partially offset by weak commodity pricing, so I guess my question is, is blended DRAM pricing up sequentially in Q2?

A - Sumit Sadana {BIO 7009258 <GO>}

Yes, I think, we don't really guide for pricing. I think we have given you a lot of data points that can help you in your modeling, but we are not going to provide guidance on the actual pricing expectation for competitive reasons, and I think the overall set of comments that you heard in the prepared remarks and the comments that Sanjay and Mark made, basically, the mix improvement continues on a good trajectory. HBM growth continues on a good trajectory. The data center overall revenue trajectory continues to be robust.

While we do have some near-term moderation in data center SSD after a lot of growth in the past quarters, the overall data center revenue trajectory remains very robust and we expect that to underpin performance through fiscal and calendar 2025. And so those are the positives and of course, we spoke about the near-term issues related to market environment, seasonality, NAND related challenges, so those are some of the headwinds.

The mix and data center growth and overall DRAM robust results is what is helping us ensure that the margin for FQ2 is still fairly similar to what we have in FQ1 in terms of 100 basis point only delta despite some of these headwinds we mentioned in some of the other areas.

Q - Harlan Sur {BIO 6539622 <GO>}

That's helpful. And then maybe to follow up, one of the -- and I've said this before, right, one of the big Micron specific inflections has been the strong increase in enterprise and data center SSD share, right? I mean, you guys have always been strong in SATA. Mainstream PCIe and NVMe is where you guys have gone from like 3% to 5% market share in calendar '21 to 10% to 12% exiting last year.

I think in Q3 your share was like 14% and it grew sequentially again in fiscal Q1. So, my guess is \$1.1 billion, \$1.2 billion of data center SSD, that's like 55% of your overall NAND business. You guys are now the number two and number three market share leader. I know enterprise SSD can be lumpy as you mentioned, right? That's impacting Q2. But do you guys expect sequential re-acceleration in data center SSD in fiscal Q3 and just overall growth in data center SSD business fiscal '25 and calendar '25 just because AI demand for those products is just so strong?

A - Sumit Sadana {BIO 7009258 <GO>}

Yes, I think you laid it out really well. We have made tremendous progress in our data center SSD business over the years. This has been an area we had been investing in for five, seven years now. And we have seen a tremendous rejuvenation in our product portfolio with industry leading

products now. So, 30 terabyte SSDs, 60 terabyte SSDs we just announced. First PCIe Gen5 60-terabyte SSD on the market, the 9550 fastest SSD in the world. So, there are lots of these amazing products that have enabled us to get to a really solid performance, several quarters of record-setting revenue and market share performance.

So, as we look ahead beyond the lumpiness you mentioned, beyond that FQ2, we do expect a growth in bit shipments restarting in FQ3 for data center SSD and then continuing from there on into future quarters. That trajectory should grow because we do expect the overall data center CapEx from our end customers to remain strong and that should enable us to get significant opportunity with the portfolio that we have that is now the strongest it ever has been in our history.

Q - Harlan Sur {BIO 6539622 <GO>}

No, that's very helpful. Thank you.

Operator

Thank you. And our next question comes from the line of Aaron Rakers from Wells Fargo. Your question, please.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yes. Thanks for doing the call back and I have two questions as well. I guess the first question Mark; I want to go back to kind of the Q&A in the prepared call. I'm trying to understand, like, you talked about constraints on gross margin into the fiscal third quarter trying to kind of frame the impact that you're thinking about from the underload. Is it 100 basis points? Is it 200 basis points? Is it less?

And when you say constraints, are you implying -- are you necessarily implying that there isn't the potential, or rather, is there the potential to see some gross margin positive drivers into fiscal 3Q relative to fiscal 2Q? Even with that underload charges, could we see gross margin flat to up? I'm just trying to understand a little bit of what you're trying to kind of frame as far as being constrained.

A - Mark Murphy {BIO 16136048 <GO>}

Yes. So, Aaron we're not guiding third quarter gross margin. So, we're just trying to be mindful to make sure that when we provide a number, it's a number that we have high conviction on. What we do know is that there are some unfavorable items that persist into the third quarter. The NAND challenging market conditions, we believe, extend into calendar Q1, which overlaps with our third quarter fiscal.

And so that's still an issue. And then, as I mentioned, in third quarter, the underload charges will begin to hit, and those will hit in the form of period costs and then just some higher cost inventory, and that effect is, you mentioned 100 basis points or 200 basis points, closer to 100 basis points impact.

And then, there are definitely some favorable effects, which continue to occur in second quarter and third quarter and beyond, and that is, the most important thing on the call, and that is, continued growth in data center, continued mix benefits from higher margin products, HBM, high capacity DIMMs, LPDRAM. And then, as Sumit mentioned, volume growth and better market conditions in NAND, but volume growth in data center SSDs continuing. So, all these items will, we believe beyond the third quarter support margin expansion.

And then, finally, Aaron, we've been clear that, we expect inventories to be up in the near term because these volume declines in NAND and DRAM in the second quarter. DIO and dollars of inventory will be up. In DRAM, we're still very tight on leading edge, and overall DRAM inventories are fine and to the point, where through the end of the year, they will continue to improve from second quarter, and then at the end of the year, overall DRAM inventories will be below our target levels, and then NAND, as that market recovers, we expect, NAND inventories, while not as healthy as DRAM, NAND inventories will improve second to third to fourth quarter.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yes, that's helpful. And then as a quick follow-up, I know a lot of focus is placed on HBM, understandably. But one of the things that you guys have highlighted over the last several quarters is LPDDR5X, and obviously getting into the Blackwell product cycle, GB200 from NVIDIA. How would you characterize your views of that ramp, and where you think Micron's relative share position is within those, that LPDDR5X data center opportunity? I'm just curious. Thank you.

A - Sumit Sadana {BIO 7009258 <GO>}

Yes, we are actually the pioneering memory supplier of LP in the data center. If you think about all of the market for LP in the data center worldwide it is pretty much Micron driving that right now. Over time there will be others, but we are in the pole position of having driven the innovations that are needed for a data center to deploy memory that was not purpose-built for a data center RAS environment. So reliability, availability, serviceability that is built into DDR5 is not available in an LP solution, and Micron has made some important innovations in that product category, and hence we are the ones, who are really partnering with customers to drive this significant opportunity for us.

And this is something that is a really important part of our overall strategy. It's an important part of our data center business. And when you combine this with the advantages that we have driven in the high-capacity DIMM portion of the data center market, where we were the first company with mono-die-based 128-gigabyte RDIMMs, then you look at the overall advantage that we offer to customers with LP in the data center, high-capacity DIMMs in the data center, these two categories for the data center in itself is going to be multi-billions of dollars.

HBM separately is multiple billions of dollars. Data center SSD is going to be multiple billions of dollars, and the combined momentum of that is what you're seeing drove the 40% sequential growth in data center, 400% year-over-year in FQ1, and the greater than 50% of our revenue now that comes from data center.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yes, thank you.

Operator

Thank you. And our next question comes from the line of Chris Caso from Wolfe Research. Your question, please.

Q - Chris Caso {BIO 4815032 <GO>}

Yes, hi. I guess the first question is maybe just some clarity on what may have precipitated some of the cautions, some of the incremental caution here, and what might have changed over the last couple of months. I know that it's kind of back and forth a little bit. There was some cautious signs kind of back in August, and, they kind of receded in the last earnings call. Was it -- I guess

the question is, is it a function of the customer inventories turned out to be a bit more than we had expected, or do you think it was a function of demand?

A - Sumit Sadana {BIO 7009258 <GO>}

Yes. Let me try to maybe address what changed. So, if you think about, over the course of CQ4, we have seen a push out of the PC refresh cycle that our customers had been anticipating, we had been anticipating. And it's not like the refresh cycle won't happen. It will happen in 2025. It's just been a little bit delayed. There are numerous drivers, which we mentioned in our prepared remarks as to why we believe and our customers believe that the refresh cycle will happen. The average age of the PC has become pretty long in the tooth, and it is strided for an upgrade.

And as people look to keep their PCs for a number of years, they will look to ultimately future-proof the hardware specs, which is why we believe that even with very modest unit volume growth, the mixed improvement that will come from higher memory content in these PCs will be a positive driver in 2025.

But the delay in that upgrade cycle means that our calendar '24 PC shipment forecast at the unit level for PCs at our end customers has been reduced and is now very flattish year-over-year in calendar '24. So, that has been one driver.

The other is that definitely the inventories that we had highlighted in the last earnings call that we expected by spring would become healthier at our customers. That inventory and inventory reduction, as well as the seasonality of CQ1, those are continuing impacts, some impact coming from the moderation. Previous discussion we just had a short while ago about lumpiness of demand on the data center SSD side, so some moderation in data center SSD into CQ1 after a significant bout of buying over several quarters in calendar '24.

So, those are the things that have impacted the near-term outlook. But I just wanted to mention that most of the impact is limited to consumer-oriented segments and the trajectory of demand in the data center continues to be very robust. Our own data center segment view of the overall revenue trajectory through fiscal '25, calendar '25 remains in a very solid trajectory. You have seen our FQ1 results as well that we highlighted.

So, it's mainly the consumer-oriented segments and some very temporary moderation of the data center SSD, which we expect will pick up again in a couple of months. And so, if you think about just these consumer-oriented segments, we do expect that by spring time, the inventories will be much healthier. And then we are back to shipment growth, because right now we are shipping to these customers at a rate that is lower than their ship-out, because they are consuming DRAM and NAND at a faster rate than they are purchasing from us and from the industry.

So, that's the effect of the inventory that we expect will be in a much healthier place by spring and then a resumption of shipments and growth in shipments for the second half of the fiscal year and then things will be much better.

And overall DRAM will be continuing to be in a much healthier place. The supply is tight. HBM continues to pressure non-HBM availability of supply. So, a large part of the issue is NAND-related and we have outlined the actions we are taking on the supply there too, decisively bring our supply and balance with the demand.

Q - Chris Caso {BIO 4815032 <GO>}

That's helpful. Thank you. I mean just a follow up to that. In terms of and I think you have been very clear on what's going on with NAND, but with DRAM and I guess this lower bit demand assumption overall for Q2. How much of that is DRAM versus NAND? And with the mid-teens growth that you now have for DRAM for the year, how back-end loaded is that given what you're seeing in Q1? Because I guess you really weren't specific in NAND versus DRAM bit to demand for -- bit growth for Q1.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Well, we --

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Yes.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes, Chris, just to be clear, we did say that DRAM bits would be down. In Q2, we said that NAND would be down meaningfully.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Okay.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

The majority of revenue decline versus the second quarter will be NAND.

A - Sumit Sadana {[BIO 7009258 <GO>](#)}

So, yes, I mean, the DRAM decline in FQ2 is largely driven by the dynamics I described on the segment side for consumer-oriented segments, right? I mean, the data center segment is very strong on the DRAM side, very strong. So, it's really the consumer-oriented segments. And, we feel that that's -- we're going to get past a lot of that headwind in the spring time frame. So, after CQ1, things will start to improve there. So, we do expect sequential increases in volume in both DRAM and NAND starting in FQ3. FQ2 is what Mark had mentioned, the sequential decline in DRAM volume, but a meaningful decline sequentially in NAND volume.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Okay. Got it. Helpful clarification. Thank you.

Operator

Thank you. And our next question comes from the line of Vijay Rakesh from Mizuho. Your question, please.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

Yes. Hi, Mark and Sumit. Thanks for doing this callback. Just wondering on the -- as your HBM capacity maybe goes up next year, how much is your conventional DRAM capacity coming down? And can you cite the HBM capacity, how much do you think that goes up next year? I have a follow-up.

A - Manish Bhatia {[BIO 18045807 <GO>](#)}

Hi, Vijay. It's Manish. So, I think, we've given, the data points that, as the -- as we go towards mature yields in HBM, that the trade ratio between HBM and DRAM is, HBM3E and our conventional DRAM is at 3. So, that's, you can sort of do the math that of how much of our as we grow our HBM share towards our natural bit share, which we expect to achieve sometime, next year in calendar '25 you can do the math and determine how much of that impact will be on our conventional DRAM wafer starts. Because we'll, essentially for everyone, we're going to have to have, three wafer starts in HBM for every one that we would have had for the same number of bits in conventional DRAM.

And we are focused on being able to maintain our DRAM bit share, as we go through the HBM transition here next year. And I think our DRAM overall bit growth will be in line with the industry bit growth numbers, demand numbers that we had talked about for calendar '25.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

Got it. And then on the CapEx side, can you talk to what's the split between HBM and conventional DRAM and NAND? Thanks.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

I don't know if we're breaking it out specifically, but we have said that, when you think about HBM in terms of both the capital that's specific to the HBM product, whether that's in the front-end fabs or in the specific assembly and test process and the clean room needed for new HBM capability, all of that. Plus, you also have the CapEx to be able to maintain the bit output as you are impacted by that trade ratio. That overall, HBM is the largest portion of our CapEx for this fiscal year. Other areas that we're focused on are in facilities and construction and also some in back-end, as you know, we have a facility that we're investing it in India and others.

Yes, Vijay, NAND is -- clearly, we've mentioned we've cut CapEx in NAND. It's lower. There is a percent spend in that is lower than percent of the business on revenue. So, it's a much lower spend. And the DRAM spend, as Manish mentioned, is largely driven by this trade ratio effect and HBM requirements there.

Q - Vijay Rakesh {[BIO 5884146 <GO>](#)}

That's very helpful. Thanks.

Operator

Thank you. And our next question comes from the line of Brian Chin from Stifel. Your question, please.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Hi there. Good afternoon. Thanks for letting us ask a few questions. Maybe first on the NAND side, we've heard of at least one other NAND supplier reducing wafer starts, maybe kind of similar as to how you've disclosed today. But back in early 2023, some of the larger suppliers were slow to take actions to reduce wafer inputs. Do you expect maybe some of these other suppliers to follow more quickly this time with similar actions or lag like they did previously?

A - Sumit Sadana {[BIO 7009258 <GO>](#)}

Difficult for us to project or predict what others would do. We'd rather not speculate others' activities. What we can say is how we look at the market and the decisive actions we are taking

immediately. And we feel like these things will at least help Micron's view of supply growth coming from us and our view of demand growth to be aligned. So, that's what we are doing.

Q - Brian Chin {BIO 5763426 <GO>}

Okay. Thanks. And then on the DRAM side, I don't think it was that many quarters ago that your DDR5 bit shipments crossed over DDR4, and now you're indicating only 10% or so of your global DRAM sales will be from DDR4 in the next several quarters through fiscal year end. That's actually a lot lower ratio than I would have expected. Does this shipment mix actually closely match your internal production mix, meaning that it's not just your holding back inventory, but structurally you've shifted a high majority of your DRAM production to DDR5, HBM, specialty, those type of products?

A - Sumit Sadana {BIO 7009258 <GO>}

Yes. I mean, that's correct. I mean, we do have -- I mean, just for clarity, DDR4 and LP4, right, that's in that 10% approximately range of revenue for the remainder of fiscal year. And we do have long life cycle products that we support for the different segments like industrial, automotive, over time with more defense, aerospace, et cetera, that will have some of this long cycle effect.

However, as our overall revenue grows, the mix of these products will remain at a very contained level. We have shifted a lot of our mix to these leading edge products, and our leading edge nodes are very constrained. So, we are dedicating a lot of capacity to a very fast ramp of HBM, and you're seeing the results from that. And even as we continue to increase our view of calendar '25 time of HBM, we are maintaining our view of getting to our natural share in the second half of calendar '25.

So, you're seeing that a lot of mix shift is happening, and I spoke about the LP leadership in the data center, DDR leadership, and the growth that is taking place. And even the growth in LP5 on the flagship smartphone side that we are supporting. So yes, definitely that shift is very meaningful.

Q - Brian Chin {BIO 5763426 <GO>}

Great. Thank you.

Operator

Thank you. And our next question comes from the line of Quinn Bolton from Needham & Company. Your question, please.

Q - Quinn Bolton {BIO 3192909 <GO>}

Hey, Mark. I just wanted to come back to your comment on the NAND and I think you said that NAND would account for most of the sequential decline. I mean, if that's right, it could be down 30% plus. So, I just want to make sure I heard you right that, that was indeed what you said, and then I've got a follow-up on the consumer-facing business.

A - Mark Murphy {BIO 16136048 <GO>}

Yes, it's the majority of -- it's the majority of the revenue decline versus the second quarter sequential.

Q - Quinn Bolton {[BIO 3192909 <GO>](#)}

Okay. Thank you. And then just you talked about the high inventory levels last quarter. It sounds like you thought you might have made some progress through fiscal Q1. Just wondering if you might be able to sort of quantify, how much is left, can you give us any sense on, are you looking to reduce a week, or two weeks, in -- through the spring? And from where you started last quarter, I assume most of that inventory reduction is happening in the second quarter that you probably didn't get through that much in fiscal Q1, but just wondering if you might be able to sort of, from where you started, how much have you gotten through and how much is left, maybe on a percentage basis if you don't want to talk on a week's basis.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Are you talking overall inventories, Quinn?

Q - Quinn Bolton {[BIO 3192909 <GO>](#)}

Yeah. Just on mostly on the consumer-facing business that seems to be kind of driving most of that sequential decline into fiscal Q2.

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes. So, we didn't break down inventories by market. We did make progress in the first quarter, as you noted. DIO was down below 150 days overall. But we have a volume-related decline or deterioration in DIO in the second quarter. So on a, we'll have DIO go up first to second quarter, we'll have absolute inventory dollars go up. And again, this is a volume-driven phenomenon.

And then from there, in part due to the continued data center strength, the resumption of data center SSD volume growth and the refresh cycle and content-driven growth in consumer and consumer-related markets, we do see our inventories drawing down through the year, both in NAND and DRAM.

Now, in DRAM, we will need our leading-edge inventories that we have now just to be able to supply the market we see. So, we're going to, on our forecast today and total DRAM inventories below our target levels by end of fiscal year.

Q - Quinn Bolton {[BIO 3192909 <GO>](#)}

Sorry, Mark. I guess my question is more inventory of the customers rather than on your balance sheet. Sort of, you said you're currently shipping below consumption in Q2. I guess maybe the other way is, it sounds like you think you start shipping in line with consumption again by Q3?

A - Mark Murphy {[BIO 16136048 <GO>](#)}

Yes, I thought that was covered earlier on that we see inventories getting to a better place in the spring, so. But as Sumit mentioned, we're shipping in less than they're shipping out or consuming. So, we have confidence that both in their demand and related our inventory consumption through fiscal year '25 beyond our second quarter.

Q - Quinn Bolton {[BIO 3192909 <GO>](#)}

Yes, perfect. Thank you.

Operator

Thank you. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Thanks. Mark, I think, I just want to clarify that I heard what you said. So you, in terms of your DRAM bit shipments, you're sort of saying that they're going to be in line with the mid-teens for calendar '25, is that right? Number one. And then number two, are you going to under-ship the market? Because I think you said demand is growing doubles, which you took that down. But I would think you're going to under-ship that.

A - Mark Murphy {[BIO 16136048](#) <GO>}

We said that industry bit supply or industry bit demand mid-teens, we gave -- we said with DRAM, yes for DRAM. And we said that volumes would be down in the second quarter and expect to ship in line with the industry.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

For the year -- full year?

A - Mark Murphy {[BIO 16136048](#) <GO>}

For the year.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Yes, okay. And then what about in NAND? I mean, NAND is -- NAND demand is up low doubles, but I would guess you're going to under-ship that for the year. Is that fair?

A - Sumit Sadana {[BIO 7009258](#) <GO>}

No, I mean, for the year, we expect to have stable bit share in calendar '24 and '25. Across both DRAM and NAND, generally that is true across '24 and '25. We expect to have relatively stable, some noise here and there, but relatively stable bit share across both of those years.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Okay. So, you'll ship, so your NAND shipments are going to be up like close to that low double-digit demand. Is that fair?

A - Sumit Sadana {[BIO 7009258](#) <GO>}

Yes, yes.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Okay. All right, and then --

A - Sumit Sadana {[BIO 7009258](#) <GO>}

There's always the calendar versus fiscal, right, I just want to be careful because that skews things somewhat, but we have a four-month skew between calendar and fiscal year, so that changes things.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Yes, but I mean, I was talking calendar to calendar, so I'm just trying to just say in the calendar year, because you're giving us a demand number for calendar year, so I'm saying in that calendar year you're going to shift in line roughly for both DRAM and NAND.

A - Sumit Sadana {BIO 7009258 <GO>}

That's right.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Got it. Okay. All right. I'll let somebody else ask.

Operator

Thank you. (Operator Instructions). Our next question comes from the line of Krish Sankar from TD Cowen. Your question, please.

Q - Hadi Orabi

Hey, guys. This is Hadi for Krish. Would like to clarify your comments regarding HBM and the volume and pricing agreements being locked in. Are these agreements cancellable? So, I wonder under what conditions can customers cancel? I'm wondering if there are specific like duration limits or restrictions in place? Thank you.

A - Sumit Sadana {BIO 7009258 <GO>}

Yes, I mean these are agreements that relate to amount of bits and cubes of HBM that our customers are going to purchase 8-high versus 12-high by quarter, with pricing associated with them. And once we get into certain relatively lengthy lead times, then we get firm orders, because of these lead times that we build with and ship for. And given the environment in the industry, more of the discussion is around chasing more supply, because we have an extraordinarily strong product that customers want more of. And we are ramping as fast as possible, fast as humanly possible to be able to grow our share to the targets that we have outlined.

And so, yes, customers can cancel if it is not in that firm time for the purchase order. If it is beyond that timeline, yes, customers can make changes. But there is a fairly lengthy timeline, when based on very long lead times that they have to place firm purchase orders for. And customers have been doing that with consistency. And, that's what gives us the confidence for the very strong ramp ahead.

Q - Hadi Orabi

Got it. Thank you.

Operator

Thank you. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi. Thank you for taking the follow-up here. I had one question on HBM and how to think about the ramp going forward on a quarterly basis. You guys do not report HBM revenue, so I am guessing a little bit here. But based on everything you have said over the past couple of quarters, we are sort of estimating you did, I don't know, \$800 million, \$900 million in revenue in the quarter. I assume it's going to be up again nicely this quarter.

So, when I think about the starting point this quarter and then the comments you made on calendar '25 TAM and your market share goals, it feels like the trajectory kind of flattens out from here, still up, but flatter. I am curious, am I thinking about it the right way? And if so, what are some of the constraints? Is it space? Is it TSV capacity, both? And when should we expect your supply side, your capacity to increase nicely? Is it the Taiwan facility? Any insights there would be really helpful. Thank you.

A - Sumit Sadana {[BIO 7009258](#) <GO>}

I will give you a couple of opening comments and then I will pass it over to Manish to talk about the second half of your question that relates to space and so on. Yes, we did give you the expectation of how we would ramp and how we get to this market share that is roughly equivalent to our DRAM supply share, which is in the low-20s and get there in the calendar second half with a view that there is going to be a \$30 billion plus TAM on HBM for calendar '25, right? So, that's sort of the outline that we have given you.

Now, you are right, we don't report HBM numbers. We may choose to provide some data points in the future, but we haven't done so except for that over \$100 million point that we had mentioned, when we first crossed that. The numbers that you outlined for FQ1 are a little bit high, but nonetheless, I think, there is going to be still a pretty robust ramp occurring on the HBM revenue for us to hit the targets that we have outlined based on the TAM expectation and the growth of TAM on a quarterly basis that we had mentioned.

And with that, I will just pass it on to Manish to answer the second half of your question.

A - Manish Bhatia {[BIO 18045807](#) <GO>}

Yeah, Toshiya. So, we're pleased with how the 8-high ramp has gone. We made some comments on the prior call about how that had exceeded our expectations in the last quarter, and we're continuing to have a strong ramp forward here. We think that the process choices we've made are -- this demonstrates that we have a scalable process and that it's going to be able -- we're going to be able to transition it to the 12-high, as we move -- transition, our capacity towards that ramp, which will become increasingly important as we go through calendar year '25.

And really, it's just a ramp. You have -- it's just a ramp of adding capacity across many different dimensions, and it's just gradually happening over time as we are, installing new equipment and qualifying that new equipment and growing our capacity week-over-week, as we move towards that goal.

Keep in mind, we had zero -- we had no product in HBM3 and very limited production in HBM2E on a process that was quite a bit different than what we have right now in HBM3E. So, we really are starting, from a very low point, as we were beginning this HBM3E ramp and are continuing to grow, and we're, adding capacity, gradually as we went through this year and continuing into next year.

So, as Sumit said, I think, we would expect that as we go through the year and as we get to the 12-high being a larger portion of our mix through the year, you will still see solid revenue growth,

as we're going quarter-over-quarter through the rest of fiscal '25 and even as we go through calendar year '25.

Q - Toshiya Hari {BIO 6770302 <GO>}

That's really helpful. Thank you so much.

Operator

Thank you. This does conclude the question-and-answer session as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.

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